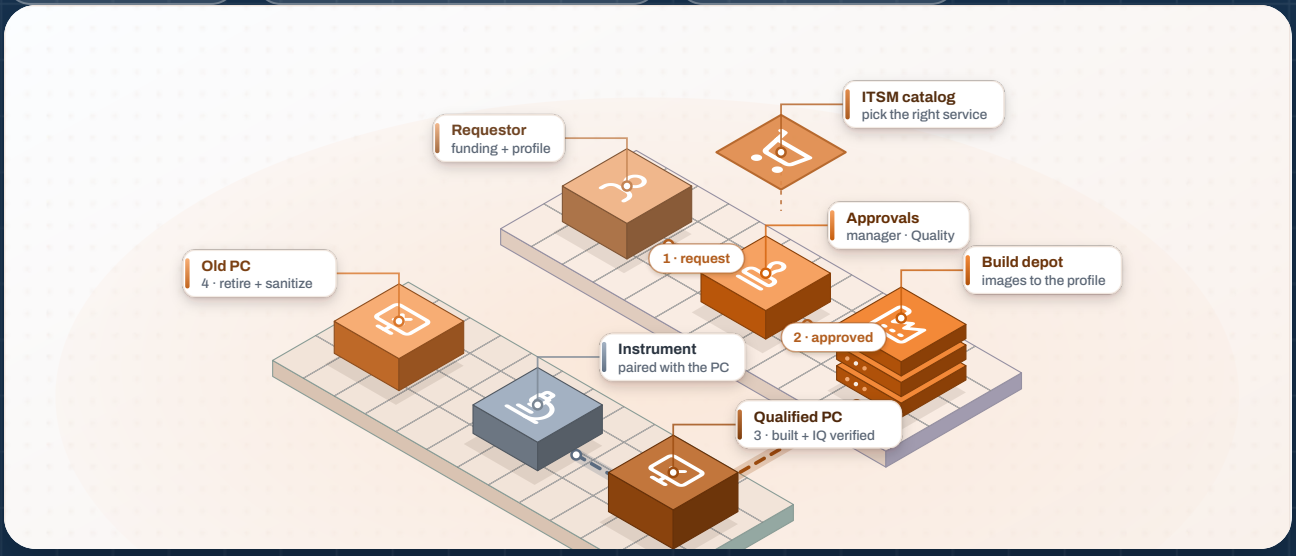
STANDARD OPERATING PROCEDURE · FIELD GUIDE SERIES

SOP-LAB-003

Requesting and Provisioning a Regulated (GxP) Lab Workstation

From funding to handover and retirement: the ITSM catalog path for a standard build, a qualified (validated) build, a break-fix rebuild and a decommission, with an inventory script and request checklist.

Regulated Lab IT | Computerized System Validation | Service Management



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LEARNING OBJECTIVES

After working through this procedure you will be able to:

- Explain why a lab PC that runs a validated (GxP) application is treated as part of a computerized system, and what that means for its build.

- Assemble everything a request needs: funding code, approvals, workstation configuration profile and a machine inventory.
- Choose between a standard build, a qualified build, a break-fix rebuild and a decommission, and follow the matching catalog path.
- Select an appropriate Long-Term Servicing Channel (LTSC) operating system and a ConfigMgr maintenance window that suit instrument work.
- Collect a workstation inventory with PowerShell (Get-CimInstance) to attach to the request.
- Retire a workstation while preserving regulated data and returning licensed hardware.

LEAD TIME	DIFFICULTY	MINIMUM ACCESS	RISK
≈ 1–3 weeks (form shows estimate)	Process-heavy, low technical risk	ITSM portal user + build portal user	Rework and delay if the profile is incomplete

CHAPTER 1

Concepts You Need First

Ordering an office laptop is a shopping exercise. Ordering a PC that will sit beside a mechanical tester, a chromatography system or a microscope in a regulated R&D lab is an engineering exercise. That PC runs instrument software, stores raw data and may produce records that support a medical device or drug submission. If it is built wrongly, patched at the wrong moment or retired carelessly, the lab can lose data, fail an audit or invalidate months of testing. This SOP explains the request-and-provision process for such a **managed lab workstation** and the reasoning behind each step.

> **KEY CONCEPT · THE PC IS PART OF THE INSTRUMENT**
>
> Think of the lab workstation the way a pilot thinks of an aircraft's avionics: you cannot swap a part without knowing exactly what is installed and proving it still works. Regulators view the workstation, the instrument software and the instrument together as one *computerized system*. Its configuration must be specified in advance, built to that specification, verified and kept under change control for its whole life.

1.1 GxP and computerized system validation

GxP is the family of "good practice" regulations (manufacturing, laboratory, clinical). When a workstation runs a *validated* GxP application locally, the organization must hold documented evidence, called **computerized system validation (CSV)**, that the system does what it is intended to do. Industry practice follows the risk-based approach of GAMP 5, often drawn as a "V": requirements on the left, verification on the right.

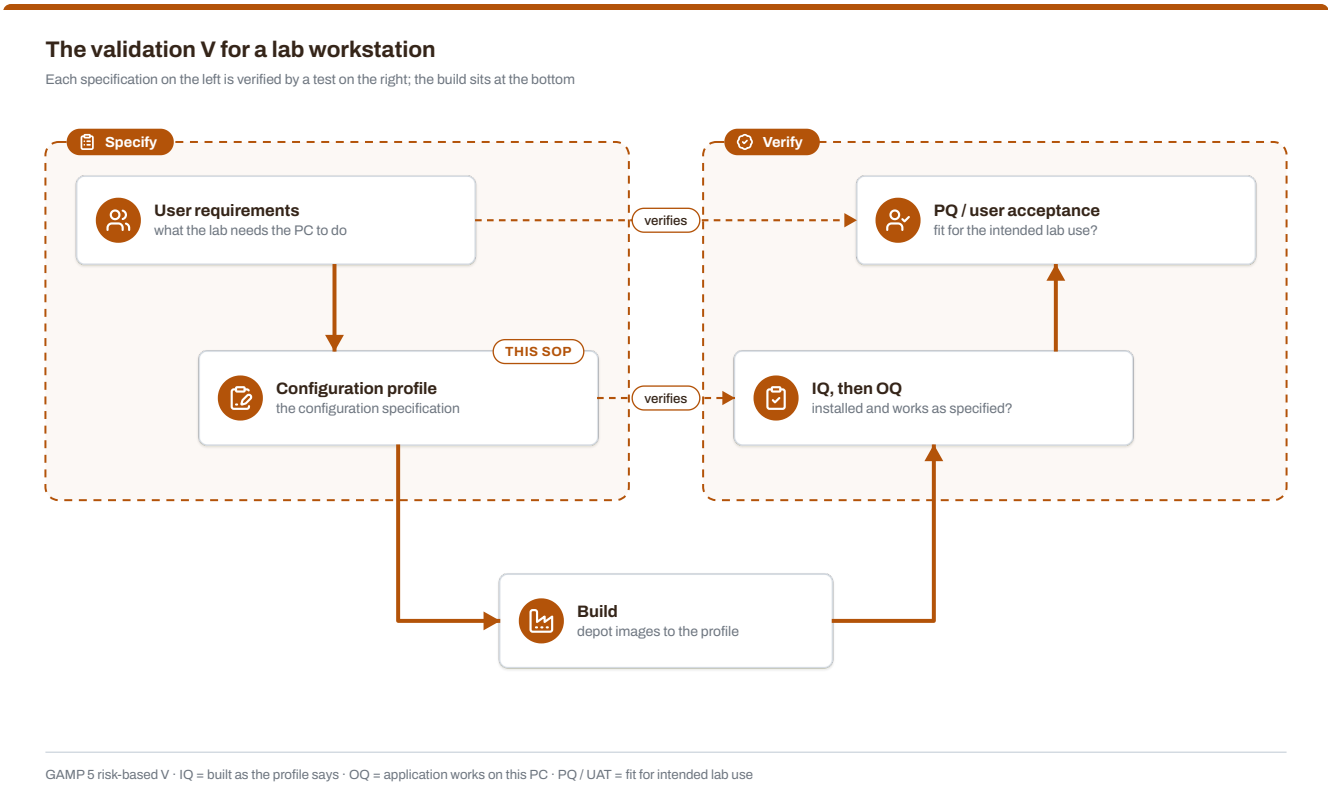


Figure 1. The configuration profile you complete in this SOP is the specification the build is later verified against during installation qualification (IQ).

Term	Meaning for the requestor
IQ	Installation qualification: the PC was built exactly as the profile says (model, OS, applications and versions, settings).
OQ	Operational qualification: the application functions correctly on this PC (for example, captures data and writes audit trail).
PQ / UAT	The lab confirms it is fit for its intended use with real workflows.
Qualified workstation	A PC whose build has been through IQ (and usually OQ). Rebuilding it means repeating that evidence, which is why a qualified rebuild is a separate service.

1.2 The workstation configuration profile

The **workstation configuration profile** (in many organizations a form in a separate build portal) is the specification for one PC. It captures who uses it, what it runs, how it is patched and who may administer it. The build depot builds from it; Quality verifies against it; and later changes are measured against it. A complete profile is the single biggest factor in a fast, right-first-time build.



Figure 2. A complete profile is the biggest single factor in a right-first-time build, and sections 2 to 5 are where incomplete requests usually stall.

1.3 Long-Term Servicing Channel (LTSC) operating systems

Normal Windows editions receive a feature update every year. Each one can change drivers, .NET behavior or UI elements that instrument software depends on, forcing re-validation. **LTSC** editions receive only security and quality updates for a long fixed life, with no feature updates, which is why they are the default for lab and instrument PCs.

Edition	Support ends	Lab guidance
Windows 10 Enterprise LTSC 2019 (1809)	January 2029	Existing validated PCs only; plan migration.
Windows 10 Enterprise LTSC 2021	January 2027	Short remaining life; avoid for new builds.
Windows 10 IoT Enterprise LTSC 2021	January 2032	Hardware-bound (OEM) licensing; useful for instrument-embedded PCs.
Windows 11 Enterprise LTSC 2024	October 2029	Preferred for new builds where the vendor supports Windows 11.
Windows 11 IoT Enterprise LTSC 2024	October 2034	Longest life; OEM licensing through the hardware supplier.

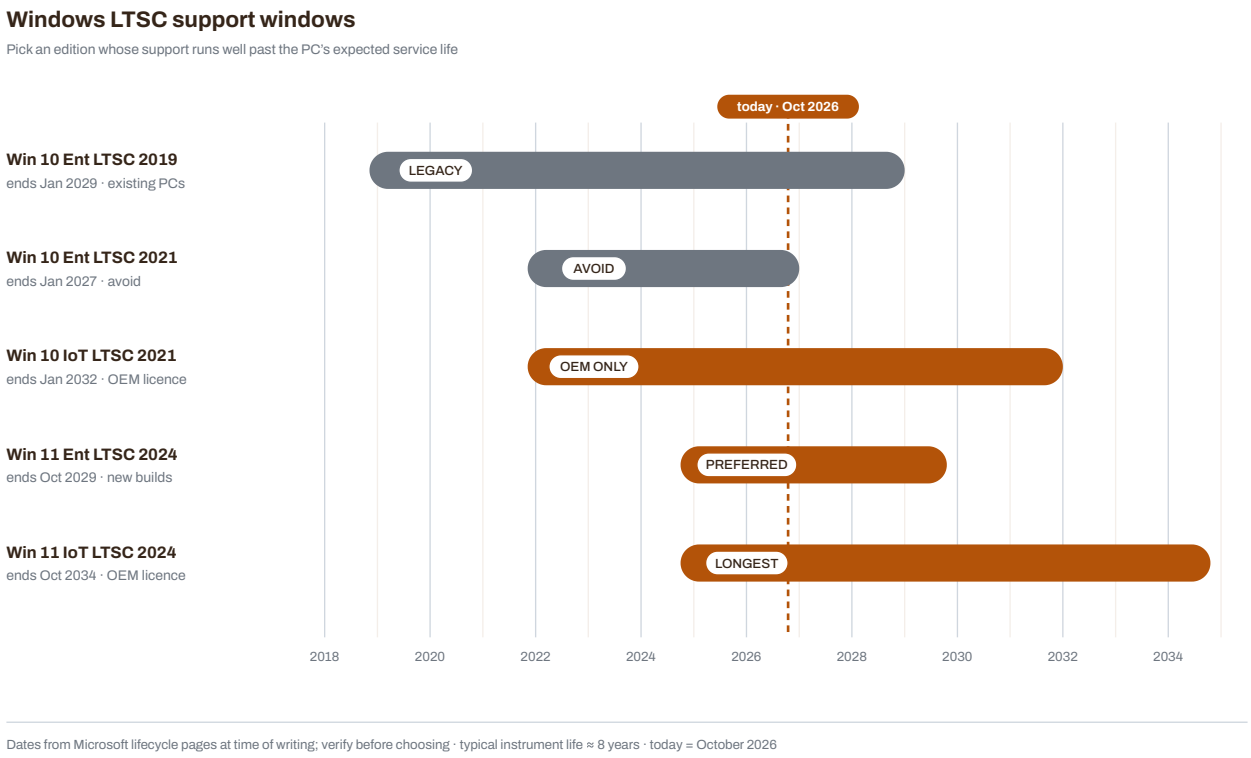


Figure 3. A PC ordered today should sit on a bar that runs well past its service life; Windows 10 Enterprise LTSC 2021 leaves barely a year, while the 2024 IoT edition covers most of a typical eight-year instrument life.

> **NOTE · VERIFY BEFORE YOU CHOOSE**
>
> Support dates come from the Microsoft lifecycle pages at the time of writing; confirm them, and confirm the instrument vendor's supported OS list, before submitting. Selecting anything other than the organization's standard LTSC/IoT edition normally requires a written justification in the profile comments.

1.4 Maintenance windows

Lab PCs must be patched (an unpatched instrument PC on the network is a classic ransomware entry point), but an automatic reboot during a 72-hour data acquisition destroys the run. Microsoft Configuration Manager (ConfigMgr, formerly SCCM) resolves this with **maintenance windows**: time ranges attached to a *device collection* during which required deployments may install and reboot. The profile asks you to pick one.

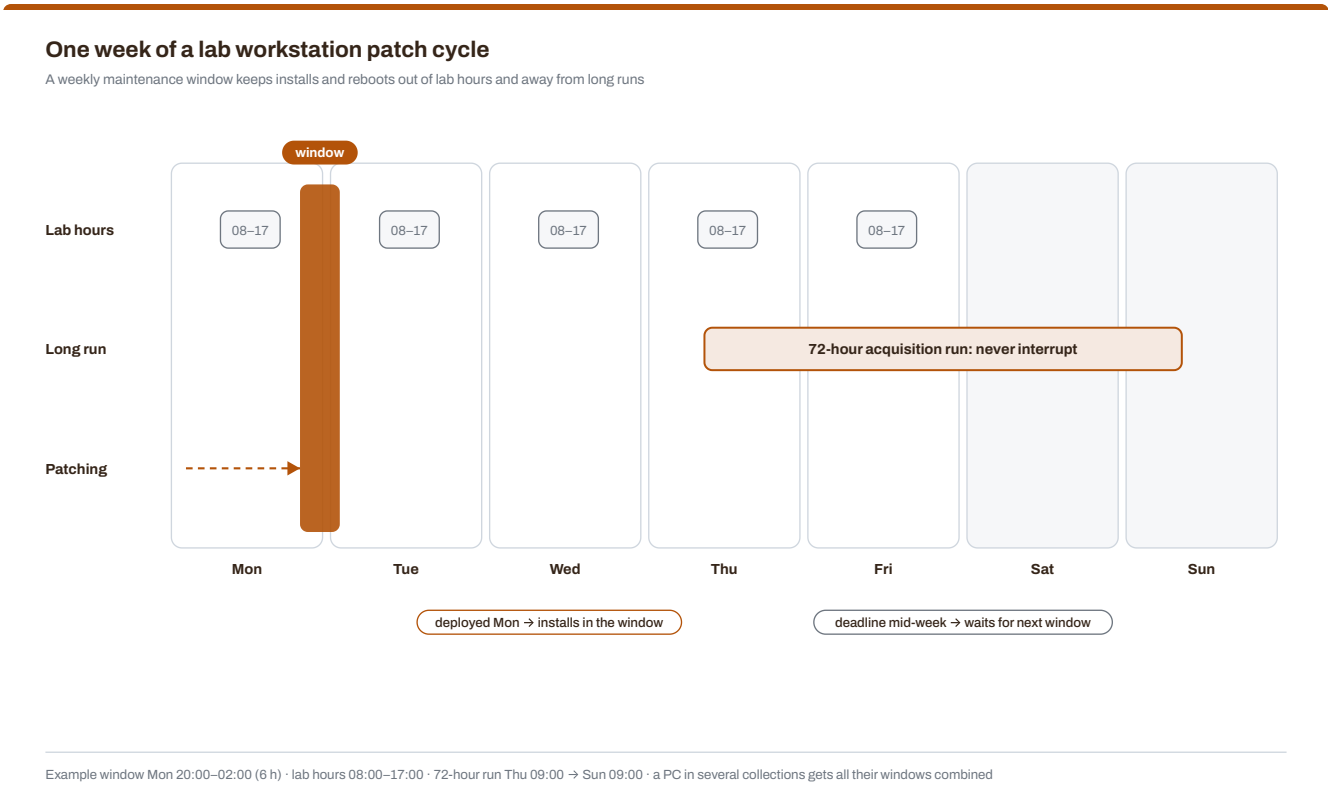


Figure 4. Patches deployed early in the week still wait for the window, so a long acquisition run is never interrupted by an install or reboot.

- A device in several collections gets the *combined* windows of all of them, so a stray collection membership can open unexpected reboot times.
- Windows can be scoped to *All deployments*, *Software updates* or *Task sequences*.
- A required deployment whose deadline passes waits for the next window unless the deployment was created with the option to install outside the window.
- Choose a schedule *with* automatic reboot unless there is a documented reason not to; updates that are installed but not activated by a reboot still leave the PC vulnerable.

1.5 Choosing a path

A · Standard build

New or replacement PC for a non-GxP or GxP use, built from the profile.

BEST FOR: MOST NEW ORDERS

B · Qualified build

Standard build plus IQ/OQ evidence and Quality approval.

BEST FOR: VALIDATED APPLICATIONS

C · Break-fix rebuild

Re-image an existing PC tied to an open incident; standard or qualified.

BEST FOR: FAILED OR COMPROMISED PCS

D · Decommission

Retire a PC with data retained and licensed hardware returned.

BEST FOR: END OF LIFE, AFTER REPLACEMENT

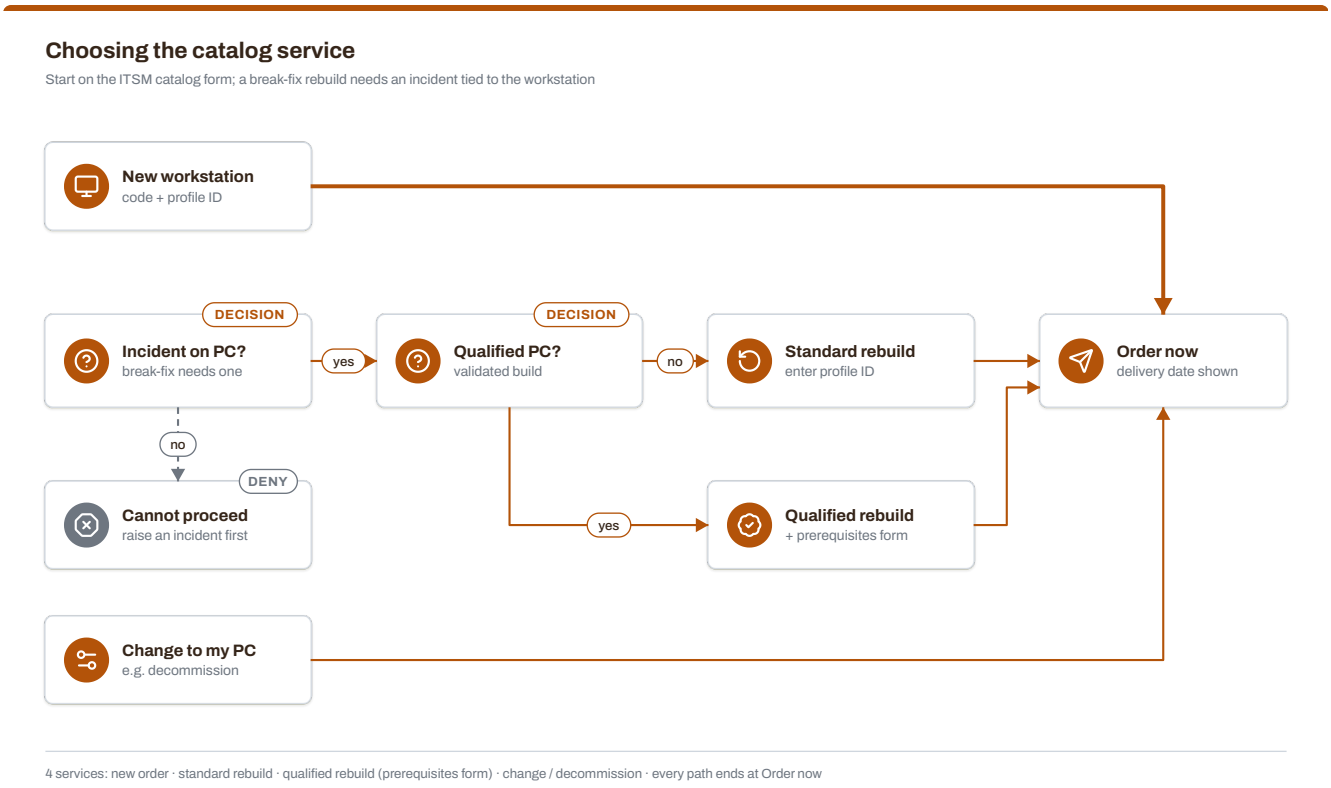


Figure 5. Pick the service before filling anything in: a break-fix rebuild with no incident on record cannot proceed.

CHAPTER 2

Access, Roles and Information to Gather

2.1 Level of access required

Task	Who / least privilege	Notes
Obtain funding code	Requestor's manager; finance contact	Follow the normal chain of command. The request cannot be submitted without it.
Complete the configuration profile	Any employee with build-portal access	The requestor may complete it on the end user's behalf. Lab IT often helps.
Submit catalog request	ITSM (for example ServiceNow) portal user	Requestor and end user can be different people.
Approve qualified builds	Quality / CSV lead for the system	Approves the prerequisites form and later the IQ/OQ evidence.
Run the inventory script	Standard user for most data locally; local admin for BitLocker/TPM and for remote CIM sessions	Remote collection uses WinRM (TCP 5985/5986) and needs admin on the target.
Grant local admin / restrict access	Directory administrators manage the AD groups named in the profile	Name *groups*, not individuals, so access survives staff changes.
Build, patch, maintenance windows	Build depot / ConfigMgr operators (Operations Administrator or a scoped custom role)	Requestors never need ConfigMgr access.

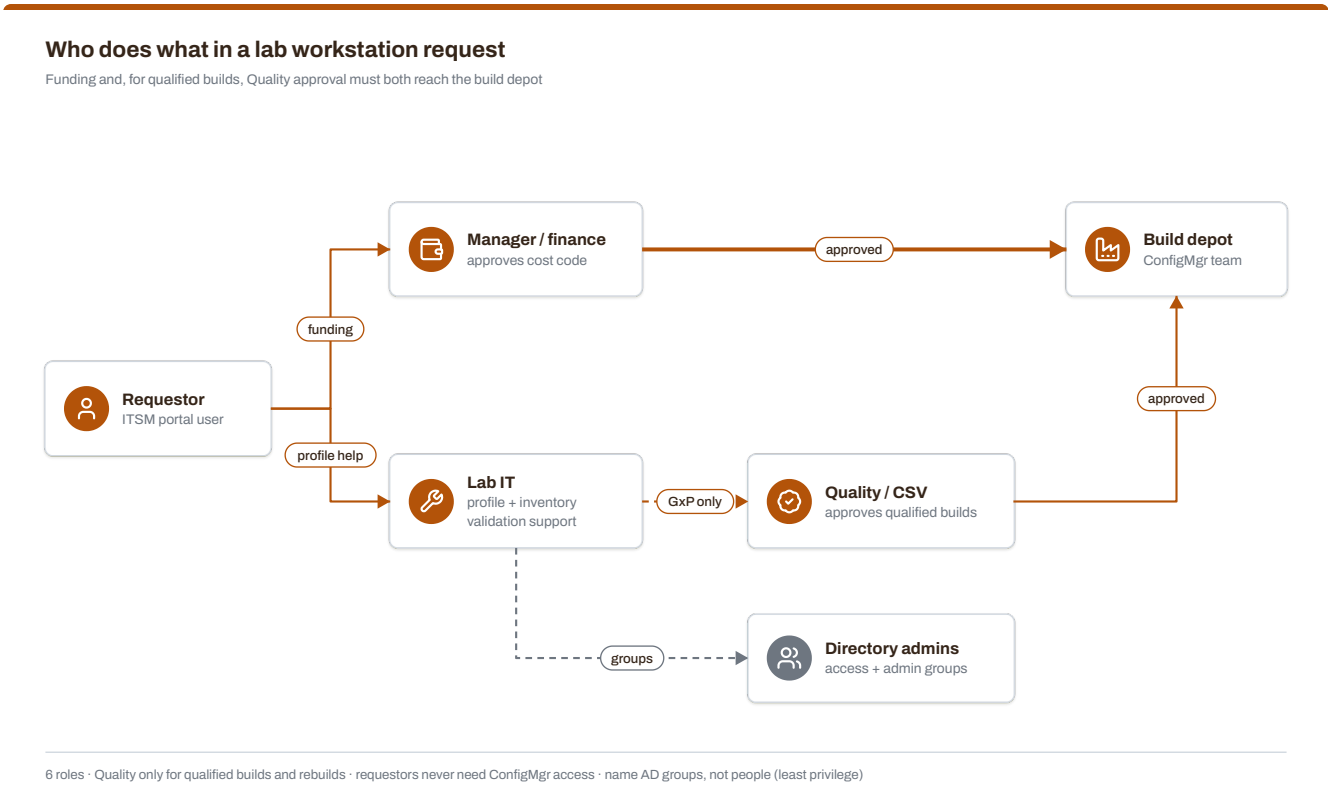


Figure 6. Quality joins the request only for qualified (validated) builds and rebuilds; everyone else follows the funding path.

> **GOOD PRACTICE · LEAST PRIVILEGE ON THE FINISHED PC**
>
> Ask for local administrator rights only if the application genuinely needs them, and grant them to a small lab-admin AD group. Every local admin on a GxP PC is someone who could change system time, delete data or disable the audit trail.

2.2 Information to gather before you start

Missing information is the most common cause of a stalled request. Collect all of this first:

- [ ] End user who will receive the workstation (name and employee ID).
- [ ] Site, lab and delivery contact (precise location goes in the request, not in public documents).
- [ ] Business justification.
- [ ] Existing workstation name(s), if this is a replacement, plus the inventory export from Chapter 3.
- [ ] Main application: name, version / service pack, and installation method (manual or ConfigMgr application IDs, with install order).
- [ ] Whether the application is validated (GxP) and whether the PC connects to an instrument (extra network ports, serial/USB adapters, acquisition cards).

- [ ] Funding / cost center code.
- [ ] Profile decisions: access groups, local admin group, screensaver lock, maintenance window, reboot behavior, antivirus exclusions.

2.3 Request checklist template

Copy this template into the request notes or keep it with the profile. It doubles as the requestor's record of what was asked for.

LAB-WORKSTATION-REQUEST-CHECKLIST.TXT TEXT

```
LAB WORKSTATION REQUEST CHECKLIST                    Change/Request: ____________
------------------------------------------------------------------------------
Request type        [ ] New   [ ] Replacement of: ____________   [ ] Rebuild   [ ] Decommission
End user            Name: _____________________  Employee ID: __________
Requestor           Name: _____________________  (if different)
Funding             Cost center / funding code: __________  Approved by: __________ Date: ______
Location            Site/lab: ____________________  Delivery contact: ____________________

GxP / validation    Validated application runs locally?  [ ] Yes  [ ] No
                    Qualified build/rebuild required?     [ ] Yes  [ ] No  (attach prerequisites form)
                    Quality / CSV contact: _____________________
Instrument          Connected to instrument? [ ] Yes [ ] No   Extra NIC ports: ___  Cards/dongles: ______

Operating system    [ ] Win 11 Enterprise LTSC 2024  [ ] Win 11 IoT Enterprise LTSC 2024  [ ] Other*
                    *Justification: ____________________________________________
Hardware            Standard model meets vendor minimums? [ ] Yes [ ] No  Model: ____________
Applications        1. Name/version: _________________ Method: [ ] Manual [ ] ConfigMgr ID: ______
                    2. Name/version: _________________ Method: [ ] Manual [ ] ConfigMgr ID: ______
                    Install order matters? [ ] No [ ] Yes: ___________________________________
Office required     [ ] Yes [ ] No      Naming convention supported by app? [ ] Yes [ ] No

Access              [ ] Restricted to group(s): _____________________  [ ] All users
Local admin         [ ] None  [ ] Group: _____________________ (justify: _________________)
Screensaver lock    [ ] Yes (GxP standard)  [ ] No - business reason: ____________________
Maintenance window  _____________________________   Reboot allowed: [ ] Yes [ ] No
DST automatic       [ ] Yes [ ] No      Image before handover: [ ] No [ ] Yes
Antivirus changes   [ ] None  [ ] Exclusions requested (attach vendor list)

Profile ID          ____________   Catalog request: ____________   Attachments: [ ] inventory [ ] prereq
form
Handover validated  Hardware [ ]  Software [ ]  Login/network/app [ ]  Data cutover [ ]   By: _____ Date:
____
```

CHAPTER 3

Collecting a Workstation Inventory with PowerShell

For a replacement or rebuild, the build depot needs to know what the current PC actually has: model, OS build, installed applications and versions, network adapters, local administrators and printers. Guessing leads to missing software on handover day. The script below collects it in one pass, locally or remotely, and writes JSON (complete record) and CSV (applications list) for attaching to the request.

KEY CONCEPT · CIM AND WMI

Windows exposes hardware and OS information through *WMI* classes such as `Win32_ComputerSystem` and `Win32_BIOS`. `Get-CimInstance` is the modern way to query them: it uses WS-Management (the same transport as PowerShell remoting) for remote machines and returns rich objects. The older `Get-WmiObject` is not available in PowerShell 7.

GET-LABWORKSTATIONINVENTORY.PS1 · BASE TEMPLATE POWERSHELL

```
<#
.SYNOPSIS  Collect a lab workstation inventory for a build / rebuild / replacement request.
.NOTES     Read-only. Local run as standard user works for most data; BitLocker/TPM and remote targets
need admin.
#>
[CmdletBinding()]
param(
  [string[]] $ComputerName = $env:COMPUTERNAME,
  [string]   $RequestRef   = 'REQ-PENDING',                # catalog request number, once known
  [string]   $OutDir       = "$env:USERPROFILE\Desktop\Inventory",
  [pscredential] $Credential
)
$ErrorActionPreference = 'Stop'
New-Item $OutDir -ItemType Directory -Force | Out-Null

foreach ($cn in $ComputerName) {
  $p = @{ ComputerName = $cn }; if ($Credential) { $p.Credential = $Credential }
  $local = $cn -eq $env:COMPUTERNAME
  $s = if ($local) { New-CimSession } else { New-CimSession @p }
  function Invoke-Here([scriptblock]$sb) { if ($local) { & $sb } else { Invoke-Command @p -ScriptBlock
$sb } }
  try {
    # ---- Hardware and OS -----------------------------------------------
    $cs   = Get-CimInstance -CimSession $s Win32_ComputerSystem
    $bios = Get-CimInstance -CimSession $s Win32_BIOS
    $os   = Get-CimInstance -CimSession $s Win32_OperatingSystem
    $cpu  = Get-CimInstance -CimSession $s Win32_Processor | Select-Object -First 1
    $disk = Get-CimInstance -CimSession $s Win32_LogicalDisk -Filter 'DriveType=3' |
            Select-Object DeviceID, @{n='SizeGB';e={[math]::Round($_.Size/1GB)}}, @{n='FreeGB';e=
{[math]::Round($_.FreeSpace/1GB)}}
    # ---- Network: physical adapters (instrument PCs often have 2+) --------
    $nics = Get-CimInstance -CimSession $s Win32_NetworkAdapter -Filter 'PhysicalAdapter=True' |
            Select-Object Name, MACAddress, NetConnectionID, Speed
    $ips  = Get-CimInstance -CimSession $s Win32_NetworkAdapterConfiguration -Filter 'IPEnabled=True' |
```

```
            Select-Object Description, IPAddress, DHCPEnabled, DefaultIPGateway
    # ---- Installed applications (64- and 32-bit uninstall keys) ------------
    $apps = Invoke-Here {
      'HKLM:\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall\*',
      'HKLM:\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall\*' |
        ForEach-Object { Get-ItemProperty $_ -ErrorAction SilentlyContinue } |
        Where-Object { $_.DisplayName -and -not $_.SystemComponent } |
        Select-Object DisplayName, DisplayVersion, Publisher, InstallDate | Sort-Object DisplayName
    }
    # ---- Local administrators, printers, USB devices ----------------------
    $admins = Invoke-Here { Get-LocalGroupMember -SID 'S-1-5-32-544' | Select-Object Name, ObjectClass,
PrincipalSource }
    $prn    = Get-CimInstance -CimSession $s Win32_Printer | Select-Object Name, DriverName, PortName,
Shared
    $usb    = Get-CimInstance -CimSession $s Win32_PnPEntity -Filter "PNPDeviceID LIKE 'USB%'" |
              Where-Object Name -notmatch 'Hub|Composite|Root' | Select-Object Name, Manufacturer
    # ---- Security state (needs admin; skipped gracefully otherwise) -------
    $tpm = try { Get-CimInstance -CimSession $s -Namespace 'root/cimv2/Security/MicrosoftTpm' -ClassName
Win32_Tpm |
                 Select-Object IsEnabled_InitialValue, SpecVersion } catch { 'not collected (requires
admin)' }

    $rec = [ordered]@{
      RequestRef = $RequestRef; Collected = (Get-Date).ToString('s'); Host = $cs.Name; Domain =
$cs.Domain
      Manufacturer = $cs.Manufacturer; Model = $cs.Model; Serial = $bios.SerialNumber; BIOS =
$bios.SMBIOSBIOSVersion
      OS = $os.Caption; Build = $os.BuildNumber; Installed = $os.InstallDate; LastBoot =
$os.LastBootUpTime
      CPU = $cpu.Name; RAM_GB = [math]::Round($cs.TotalPhysicalMemory/1GB); Disks = $disk
      PhysicalNICs = $nics; IPConfig = $ips; LocalAdmins = $admins; Printers = $prn; USBDevices = $usb;
TPM = $tpm
      AppCount = @($apps).Count
    }
    $base = Join-Path $OutDir "$($cs.Name)-$RequestRef"
    $rec  | ConvertTo-Json -Depth 4 | Out-File "$base-inventory.json" -Encoding utf8
    $apps | Export-Csv "$base-apps.csv" -NoTypeInformation
    Write-Host "Inventory written: $base-*.{json,csv}" -ForegroundColor Green
  } finally { Remove-CimSession $s }
}
```

EXAMPLE · RUNNING THE SCRIPT

INVOCATION — POWERSHELL

```
#  Local, on the PC being replaced
.\Get-LabWorkstationInventory.ps1 -RequestRef 'REQ0012345'
#  Remote, from an admin workstation, for two PCs
.\Get-LabWorkstationInventory.ps1 -ComputerName PC-LAB-0001,PC-LAB-0002 -Credential (Get-
Credential) -RequestRef 'REQ0012345'
```

Attach the JSON and CSV to the catalog request. If the request goes to an external build depot, remove serial numbers and IP addresses that your data-handling policy does not allow to leave the organization.

NOTE · THINGS A SCRIPT CANNOT SEE

Hardware license dongles, acquisition cards that need vendor drivers, instrument calibration files, and local data folders outside the user profile. Walk the PC with the lab user and add these to the checklist by hand.

CHAPTER 4

Method A · Standard Build Through the Catalog

The standard path has six steps. Each produces what the next needs: the funding code pays for the request, and the profile ID links the build configuration to the catalog order.

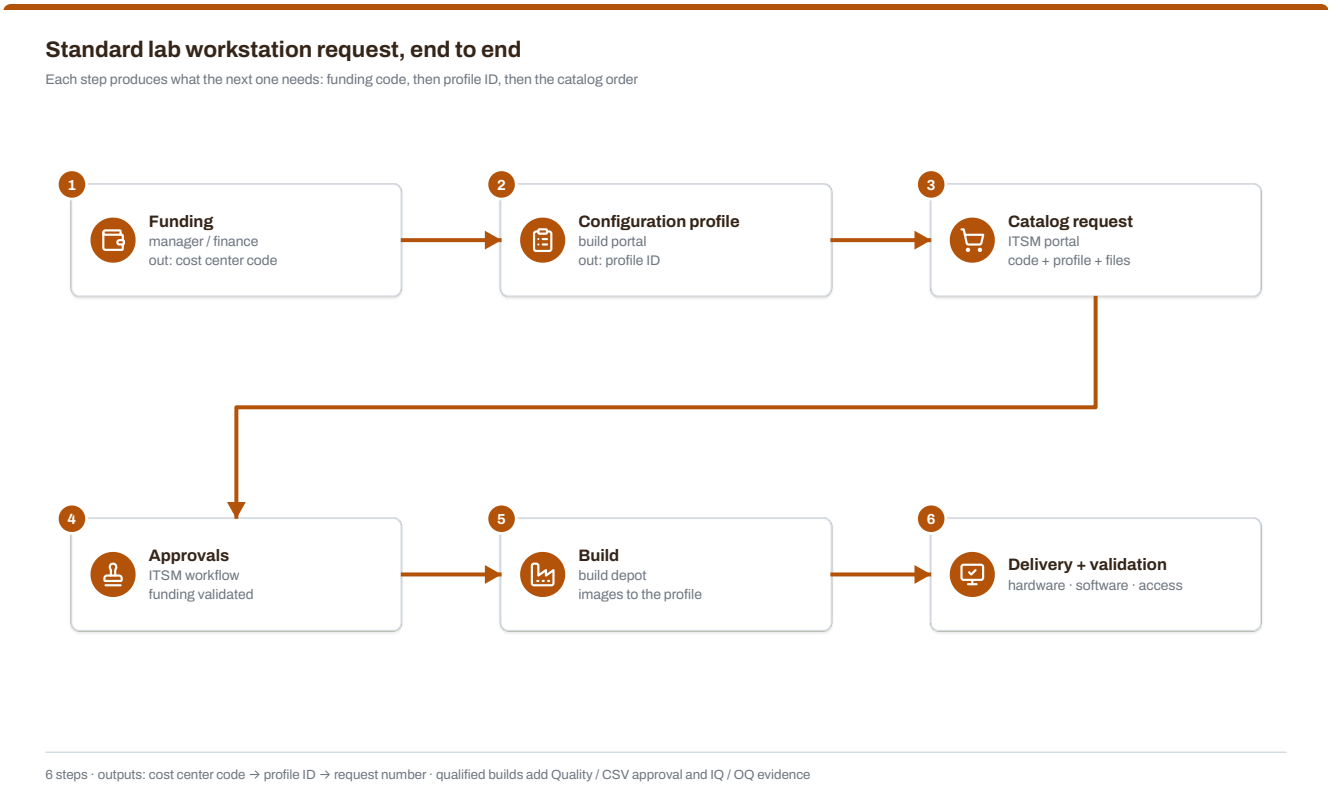


Figure 7. Each step hands the next one a concrete output, so a missing funding code or profile ID stops the request early rather than at the build depot.

STEP 1 Obtain funding approval

1. Ask your direct manager for the cost center / funding code. If they are unsure, they engage their leadership or finance contact.
2. Record the code and the approver on the checklist. The request cannot proceed until funding is identified and approved.

STEP 2 Complete the workstation configuration profile

Open the build portal and start a new profile. Use **Save for later** to keep progress and the **Comments** panel for anything the build team must know.

1. **Request type**: new or replacement lab workstation.
2. **Common information**: end user; business unit; workstation type *Laboratory*; *Will the PC run a validated (GxP) application locally?* (this drives everything downstream); main application name and version; automated installation qualification (or None); instrument connection; replaced PC name(s); site, lab and delivery contact. **Regional settings**: OS edition (LTSC, see Chapter 1), language, keyboard, time zone, and the organizational unit (OU) container the PC will be placed in.
3. **Hardware**: confirm the standard model meets the application vendor's minimum requirements, choose the form factor, and state whether extra network ports are needed (common when the instrument has its own private link).
4. **Software**: Office required or not; confirm the application tolerates the corporate naming convention; who holds the installation media and procedure (for example the business unit's application owner); and the installation method. For ConfigMgr, list each application by its ID and friendly name with `+ Add item`, and state the install order in Comments if it matters.
5. **Additional information**: access restriction (named groups or all users); local admin group; forced password-protected screensaver (*Yes* is the GxP standard; *No* needs a documented business reason such as protecting unattended acquisitions); automatic daylight-saving time; disk image before handover (default No); the maintenance window; whether updates may reboot; any antivirus change (exclusions for acquisition folders are common and must come from the vendor).
6. **Submit**: review every section, submit, and record the profile ID.

WHY IT MATTERS · DAYLIGHT-SAVING AND TIME ZONES

Audit trails are time-stamped. A PC that changes clock automatically while an overnight run straddles the change can produce records with an apparent one-hour gap or overlap. Some labs disable automatic DST on acquisition PCs and record time in UTC; decide deliberately and document it.

STEP 3 Submit the catalog request

1. Open the lab workstation request form in the ITSM / ServiceNow request portal and select the service *New workstation order*.

2. Complete requestor, end user, additional location detail and business justification / funding code.
3. Select who manages lab workstation requests at your site (internal team or a managed-service support provider).
4. Enter the **profile ID** from Step 2; this links the build to the order. Add the response date (informational) and attach the inventory files and checklist.
5. Review entries against the profile. Note the expected delivery time shown on the form, then `Order Now` (or `Save as Draft` to hold it).

STEPS 4–6 Approval, build, delivery and validation

1. **Approval**: the workflow validates funding and collects other approvals. Answer questions from approvers promptly; requests wait silently otherwise.
2. **Build**: the build depot reviews the request and profile, images the PC with the selected OS, places it in the right OU and collections (including the maintenance-window collection), installs applications in order, applies security settings and runs its build checks.
3. **Delivery and validation**: verify hardware (correct model and accessories), software (every application present at the right version and working) and access (end-user login, network, application access). For replacements, confirm data and peripheral cutover from the old PC before decommissioning it.

HANDOVER SPOT-CHECKS (ON THE NEW PC) POWERSHELL

```powershell
Get-CimInstance Win32_OperatingSystem | Select Caption, Version, BuildNumber      # expected LTSC build
Get-LocalGroupMember -SID 'S-1-5-32-544'                                          # only approved admin group
Get-ItemProperty 'HKCU:\Control Panel\Desktop' | Select ScreenSaveActive, ScreenSaverIsSecure, ScreenSaveTimeOut
Get-CimInstance -Namespace root\ccm\ClientSDK -ClassName CCM_ServiceWindow |      # ConfigMgr maintenance windows
  Select StartTime, EndTime, Type
gpresult /r /scope computer | Select-String 'Applied Group Policy Objects' -Context 0,15
```

Run the `CCM_ServiceWindow` query and `gpresult` from an elevated prompt. Screensaver settings enforced by policy may appear under `HKCU:\Software\Policies\Microsoft\Windows\Control Panel\Desktop` instead.

CHAPTER 5

Method B · Qualified (Validated) Build

When the answer to "will the PC run a validated GxP application locally?" is *Yes*, the build becomes part of a validated system. The catalog steps are the same as Method A, with these additions:

1. **Before ordering**: agree with Quality / CSV which qualification applies (automated installation qualification offered by the build team, a manual IQ protocol, or both) and select it in the profile.
2. Complete the organization's **qualification prerequisites form** (a controlled template listing application versions, settings, test scope and approvers) and attach it to the request.
3. In the profile, keep the GxP defaults: password-protected screensaver, restricted access, minimal local admins, and an automatic-reboot maintenance window outside lab hours.
4. **During the build**: the depot records an IQ: hardware model and serial, OS edition and build, each application and version, security baseline, and evidence (screenshots or reports).
5. **After delivery**: the lab or validation team executes OQ (and PQ/UAT if required) against the application. Deviations are recorded and resolved before the PC goes into GxP use.
6. Quality approves the qualification summary. The PC's configuration item in the CMDB is marked *Qualified*, and from now on every change goes through change control.

WATCH OUT · CHANGING A QUALIFIED PC

Installing "just one more tool", granting an extra admin, or moving the PC to a different maintenance-window collection are all changes to a validated system. Use the *Request a change to my workstation* service so the impact is assessed, rather than doing it ad hoc.

Aspect	Standard build	Qualified build
Specification	Configuration profile	Profile + prerequisites form + requirements
Evidence	Build checklist	IQ record, OQ scripts, summary report
Approvals	Funding	Funding + Quality / CSV
Lead time	Form estimate	Form estimate + qualification time
Later changes	Normal change process	Change control with impact assessment

CHAPTER 6

Methods C and D · Rebuild and Decommission

6.1 Method C · Break-fix rebuild

A rebuild re-images an existing workstation: after disk failure, malware, OS corruption or an unrecoverable application issue. Because it destroys the current state, the catalog requires a reason on record.

1. Make sure an **incident** exists and is linked to the workstation's configuration item. If there is no related incident, the break-fix service cannot proceed: raise one first or choose a different service.
2. Run the inventory script (Chapter 3) if the PC still boots, and back up local data to the retained network share. Record the share location in the incident.
3. Check that the existing profile is still accurate. If anything changed (new application version, new admin group), update the profile first, because the depot rebuilds from the profile, not from the old disk.
4. In the catalog form select *Request a break-fix (re)build*, then *(Re)build a workstation* or *(Re)build a qualified workstation*.
5. Select the workstation configuration item, the related incident and the profile ID. For a qualified rebuild, attach the completed prerequisites form.
6. After the rebuild, repeat the Method A delivery checks; for a qualified PC, repeat the IQ and the OQ scope Quality specifies.

DANGER · SECURITY INCIDENTS

If the rebuild is because of suspected malware, do not copy data back until the security team has cleared it, and keep the original disk (or a forensic image) if the security team asks. Restoring infected data to a freshly validated PC defeats the purpose.

6.2 Method D · Decommissioning with data retention

A lab PC is retired when a replacement has been delivered and validated, when the instrument is retired, or at hardware end of life. Its disk may hold *original* GxP records that must be retained for years, and it may hold licensed hardware worth more than the PC.

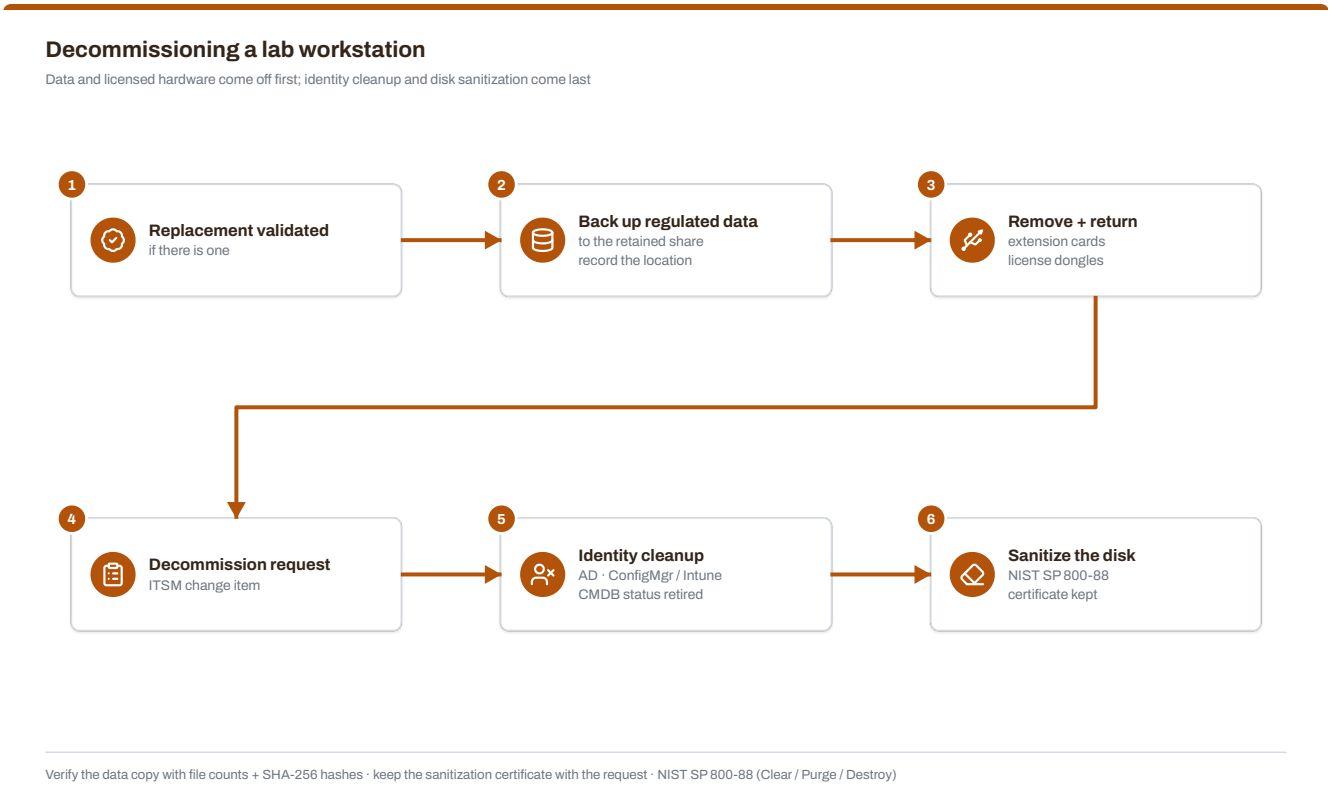


Figure 8. Retained data and licensed hardware come off the PC before the request is raised, because identity cleanup and disk sanitization cannot be undone.

1. Confirm the replacement (if any) passed delivery validation.
2. Back up or migrate all required data to a network share covered by the records retention schedule. Verify the copy (file counts and hashes) and record the share location.
3. Remove extension cards (acquisition, GPIB, serial) and hardware license keys and return them to the business unit.
4. In the catalog form select *Request a change to my workstation* › *Decommission a workstation*. Select the workstation CI and list everything removed or returned: data (with share location), cards and keys. Add the response date.
5. IT removes the computer object from Active Directory, ConfigMgr and Intune, retires the CMDB record, and revokes any device certificates.
6. The disk is sanitized according to NIST SP 800-88 (Clear, Purge or Destroy, depending on reuse) and the certificate of sanitization is kept with the request.
7. For a qualified PC, Quality may require a short system retirement record referencing where the data now lives.

VERIFY THE DATA COPY BEFORE RETIREMENT POWERSHELL

```
$src = 'D:\InstrumentData'; $dst = '\\SRV-FILE-01\LabArchive\PC-LAB-0001'
robocopy $src $dst /E /COPY:DAT /DCOPY:T /R:2 /W:5 /LOG:"$env:TEMP\archive.log"   # keeps original
timestamps
$a = Get-ChildItem $src -Recurse -File | Get-FileHash -Algorithm SHA256
$b = Get-ChildItem $dst -Recurse -File | Get-FileHash -Algorithm SHA256
Compare-Object ($a.Hash) ($b.Hash)      # no output = every file arrived intact
```

CHAPTER 7

Verification, Troubleshooting and Lessons Learned

7.1 Handover verification

Hardware and OS

- ☐ Model, RAM, disks and NIC count match the profile
- ☐ OS edition and build are the LTSC selected
- ☐ Computer name follows convention; correct OU
- ☐ Maintenance-window collection correct

Software and access

- ☐ Every application present, correct version, launches
- ☐ Instrument connects and acquires a test result
- ☐ Only approved users/groups can sign in; admins as specified
- ☐ Screensaver lock and AV settings as specified

7.2 Troubleshooting

Symptom	Likely cause and fix
Request sits in approval for days	Funding code invalid or approver unaware. Check the approval tab and contact the approver; answer information requests promptly.
Break-fix service cannot be submitted	No incident linked to the workstation CI. Raise one, link it, then retry.
Delivered PC missing an application	Not listed in the profile, or ConfigMgr ID wrong. Update the profile and request a change; do not install it ad hoc on a qualified PC.
Application fails after install order issue	Dependencies (runtime, driver) installed after the application. State the order in the profile comments.
PC reboots during a run	Wrong or extra maintenance-window collection, or a deployment allowed outside the window. Check `CCM_ServiceWindow` on the client and collection memberships.
Application cannot reach instrument	Second NIC not requested or not configured; instrument link needs a static address on a private subnet (for example 192.168.100.x).
Application rejects the computer name	Legacy software with name-length or character limits. Confirm in Step 2 and choose a naming basis it supports.

7.3 Lessons from the reference engagement

- Most delays came from incomplete profiles: missing install order, missing extra-NIC requirement, or no justification for a non-standard OS.
- Filling the checklist with the lab user, at the instrument, caught dongles and acquisition cards that no script detects.
- Linking the profile ID in the catalog request was the single step most often forgotten; the depot cannot start without it.

CHAPTER 8

Summary, Glossary and Review

8.1 Chapter summary

- A lab PC running validated software is part of a computerized system: specify it (profile), build it to spec, verify it (IQ/OQ) and control changes.
- Funding code → configuration profile → catalog request (with profile ID) → approval → build → delivery validation.
- Choose LTSC editions to avoid feature-update re-validation, and maintenance windows that protect lab hours.
- Rebuilds need an incident; qualified rebuilds need the prerequisites form and repeat qualification.
- Decommission only after data is retained and verified and licensed hardware is returned; sanitize the disk and keep the certificate.

8.2 Glossary

Computerized system	Hardware, software, people and procedures that together perform a regulated function.
CSV	Computer system validation: documented evidence that a system is fit for its intended use.
GAMP 5	Industry guide for a risk-based approach to validating GxP computerized systems.
Configuration profile	The specification of one workstation's hardware, software and settings, used to build and verify it.
LTSC	Long-Term Servicing Channel: Windows edition with security updates only, no feature updates.
Maintenance window	ConfigMgr collection setting defining when required deployments may install and reboot.
CI / CMDB	Configuration item and the configuration management database that records it.
NIST SP 800-88	US guideline for media sanitization (Clear, Purge, Destroy).

Check your understanding

1. Why does a "Yes" to "runs a validated application locally" change how the PC is built and changed?
2. What links the configuration profile to the catalog order, and what happens if it is missing?
3. Why are LTSC editions preferred for instrument PCs, and why would you avoid Windows 10 Enterprise LTSC 2021 for a new build today?
4. A PC rebooted in the middle of a weekend acquisition. Name two ConfigMgr causes and how you would check them.
5. List the steps that must happen before a decommission request is submitted.